Time Series — Solutions

Practice Exam — Week 1: Stationarity, Autocovariance & Ergodicity

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Problem 1 (5 points)

Statement.

(a) *State the definition of* weak (second-order) stationarity *and of* strict stationarity *for a process* $\{X_t\}_{t\in\mathbb{Z}}$. *(2 pts)*

(b) *Let* $Y_t \overset{iid}{\sim} \mathcal{N}(0, \sigma_Y^2)$ *and let* a, b, c *be constants. For each process, decide whether it is weakly and/or strictly stationary, and give its mean and ACVS:*

(i) $X_t = a + bY_t + cY_{t-1}$;

(ii) $X_t = Y_0 \cos(ct)$.

(2 pts)

(c) *Give a process that is strictly stationary but* not *weakly stationary, and explain why.* *(1 pt)*

Solution.

(a) $\{X_t\}$ is *weakly (second-order) stationary* if all first- and second-order moments are finite and shift-invariant: $\mathbb{E}[X_t] = \mu$ (constant), $\mathrm{Var}(X_t) = \sigma^2 < \infty$ (constant), and $\mathrm{Cov}(X_t, X_{t+\tau})$ depends on the lag τ only (not on t). It is *strictly stationary* if for every n and all $t_1, \dots, t_n$, the joint law of $(X_{t_1}, \dots, X_{t_n})$ equals that of $(X_{t_1+\tau}, \dots, X_{t_n+\tau})$ for all τ.

(b)(i) $\mathbb{E}[X_t] = a$. By independence of the Y_t,

$$\gamma_\tau = \begin{cases} (b^2 + c^2)\sigma_Y^2, & \tau = 0, \\ bc\,\sigma_Y^2, & \tau = \pm 1, \\ 0, & |\tau| \ge 2. \end{cases}$$

Depends on τ only and is finite $\Rightarrow$ weakly stationary; a linear combination of jointly Gaussian variables is Gaussian $\Rightarrow$ also strictly stationary.

(b)(ii) $\mathbb{E}[X_t] = 0$ but $\mathrm{Cov}(X_t, X_{t+\tau}) = \sigma_Y^2 \cos(ct)\cos\big(c(t+\tau)\big)$ *depends on* t $\Rightarrow$ *not* stationary (neither weakly nor strictly).

(c) An i.i.d. sequence of Cauchy (Student-t_1) variables: i.i.d. hence strictly stationary, but it has no finite mean/variance, so the finite-variance requirement of weak stationarity fails. Thus strict $\not\Rightarrow$ weak.

Problem 2 (5 points)

Statement.

Let $\{X_t\}$ *be a second-order stationary process with ACVS* γ_τ, independent *of the white noise* $\{\varepsilon_t\}$.

(a) *Determine the ACVS of* $Z_t = X_t + \varepsilon_t$. *(2 pts)*

(b) *Let* $W_t = \theta_1\varepsilon_t + \theta_2\varepsilon_{t-1}$. *Compute its ACVS* γ_τ^W *for all* τ *and its autocorrelation* ρ_1, *and prove that* $|\rho_1| \le \frac{1}{2}$. *(3 pts)*

Solution.

(a) Independence kills all cross-covariances $\mathrm{Cov}(X_s, \varepsilon_u) = 0$, and the white noise contributes variance only at lag 0:

$$\gamma_\tau^Z = \gamma_\tau + \sigma^2\,\mathbf{1}_{\{\tau=0\}}.$$

(b) By bilinearity and $\mathrm{Cov}(\varepsilon_s, \varepsilon_u) = \sigma^2 \mathbf{1}_{\{s=u\}}$ (only equal noise indices survive),

$$\gamma_\tau^W = \begin{cases} (\theta_1^2 + \theta_2^2)\sigma^2, & \tau = 0, \\ \theta_1\theta_2\,\sigma^2, & \tau = \pm 1, \\ 0, & |\tau| \geq 2, \end{cases} \qquad \rho_1 = \frac{\theta_1\theta_2}{\theta_1^2 + \theta_2^2}.$$

By AM–GM, $|\theta_1\theta_2| \leq \frac{1}{2}(\theta_1^2 + \theta_2^2)$, hence $|\rho_1| \leq \frac{1}{2}$, with equality iff $|\theta_1| = |\theta_2|$. (No process of this MA(1)-type can exceed $|\rho_1| = \frac{1}{2}$.)

Problem 3 (5 points)

Statement.

(a) *Prove that the autocovariance sequence $\{\gamma_\tau\}$ of a stationary process is* positive semi-definite*: for every $n \geq 1$ and all $a_1, \ldots, a_n \in \mathbb{R}$,*

$$\sum_{j=1}^{n}\sum_{k=1}^{n} a_j a_k\, \gamma_{j-k} \;\geq\; 0.$$

(2 pts)

(b) *Deduce the symmetry $\gamma_{-\tau} = \gamma_\tau$ and the bound $|\gamma_\tau| \leq \gamma_0$.* *(1 pt)*

(c) *Is the sequence $\gamma_0 = 1,\ \gamma_{\pm 1} = 0.8,\ \gamma_\tau = 0\ (|\tau| \geq 2)$ a valid ACVS of some stationary process? Justify. (2 pts)*

Solution.

(a) Let $W = \sum_{j=1}^{n} a_j X_j$. A variance is non-negative, so

$$0 \leq \mathrm{Var}(W) = \sum_{j=1}^{n}\sum_{k=1}^{n} a_j a_k\, \mathrm{Cov}(X_j, X_k) = \sum_{j=1}^{n}\sum_{k=1}^{n} a_j a_k\, \gamma_{j-k}.$$

(b) Symmetry: $\gamma_{-\tau} = \mathrm{Cov}(X_t, X_{t-\tau}) = \mathrm{Cov}(X_{t-\tau}, X_t) = \gamma_\tau$ (real case). Bound: by Cauchy–Schwarz $|\gamma_\tau| = |\mathrm{Cov}(X_t, X_{t+\tau})| \leq \sqrt{\mathrm{Var}(X_t)\,\mathrm{Var}(X_{t+\tau})} = \gamma_0$.

(c) No. A valid ACVS must be positive semi-definite, equivalently have a non-negative spectral density. Here

$$\mathcal{S}(f) = \sum_\tau \gamma_\tau e^{-2\pi i f \tau} = 1 + 1.6\cos(2\pi f),$$

which equals $1 - 1.6 = -0.6 < 0$ at $f = \frac{1}{2}$. The spectral density goes negative $\Rightarrow$ the sequence is not p.s.d. $\Rightarrow$ not a valid ACVS. (Already $|\rho_1| = 0.8 > \frac{1}{2}$ is impossible for an MA(1).)

Problem 4 (5 points)

Statement.

For observations $X_1, \ldots, X_n$ recall the two sample-autocovariance estimators (for $|\tau| \leq n-1$, and 0 otherwise):

$$\tilde{\gamma}_\tau = \frac{1}{n - |\tau|}\sum_{t=1}^{n-|\tau|}(X_t - \bar{X})(X_{t+|\tau|} - \bar{X}), \qquad \hat{\gamma}_\tau = \frac{1}{n}\sum_{t=1}^{n-|\tau|}(X_t - \bar{X})(X_{t+|\tau|} - \bar{X}).$$

(a) *Which estimator is biased? Which one is positive semi-definite?* *(1 pt)*

(b) *For Gaussian white noise (known mean) one has, for $\tau \neq 0$, $\mathrm{Var}(\hat{\gamma}_\tau) = \frac{n-|\tau|}{n^2}\sigma^4$ and $\mathrm{Var}(\tilde{\gamma}_\tau) = \frac{1}{n-|\tau|}\sigma^4$. Which estimator has the smaller mean-squared error, and why?* *(2 pts)* *(c) Show that if $\sum_\tau |\gamma_\tau| < \infty$ then* $\mathrm{Var}(\bar{X}) \to 0$ *as $n \to \infty$, so $\bar{X}$ is a consistent (mean-ergodic) estimator of $\mu = \mathbb{E}[X_t]$.* *(2 pts)*

Solution.

(a) $\tilde{\gamma}_\tau$ is *unbiased* (known mean) but not always positive semi-definite; $\hat{\gamma}_\tau$ is *biased* ($\mathbb{E}[\hat{\gamma}_\tau] = \frac{n-|\tau|}{n}\gamma_\tau$) but positive semi-definite — which is why it is the default.

(b) For white noise both are unbiased, so MSE = Var. Now

$$\mathrm{Var}(\hat{\gamma}_\tau) = \frac{n-|\tau|}{n^2}\sigma^4 = \left(\tfrac{n-|\tau|}{n}\right)^2 \frac{\sigma^4}{n-|\tau|} = \left(\tfrac{n-|\tau|}{n}\right)^2 \mathrm{Var}(\tilde{\gamma}_\tau) \le \mathrm{Var}(\tilde{\gamma}_\tau).$$

So the *biased* estimator $\hat{\gamma}_\tau$ has the smaller MSE. (Lesson: the goal is low MSE and positive semi-definiteness, not unbiasedness.)

(c) $\mathbb{E}[\bar{X}] = \mu$, and grouping by lag,

$$\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\sum_{i,j=1}^{n} \gamma_{j-i} = \frac{1}{n}\sum_{\tau=-(n-1)}^{n-1} \left(1 - \tfrac{|\tau|}{n}\right)\gamma_\tau.$$

By Cesàro summability, $\sum_\tau \left(1 - \frac{|\tau|}{n}\right)\gamma_\tau \to \sum_{\tau=-\infty}^{\infty} \gamma_\tau =: C(\gamma)$, finite since $\sum_\tau |\gamma_\tau| < \infty$. Hence $n\,\mathrm{Var}(\bar{X}) \to C(\gamma)$, i.e. $\mathrm{Var}(\bar{X}) = O(1/n) \to 0$, so $\bar{X} \to \mu$ in mean square, therefore in probability: $\bar{X}$ is consistent (the process is mean-ergodic).